

Modal Logic: Tableaux Examples

2026-02-18

Example 10: $\Diamond \Box A \rightarrow A$ in KTB

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All the rules of **KT** still hold in **KTb**.

The only difference is that there are now some new rules telling us that we can apply True $\Box$ sentences and False $\Diamond$ sentences back up the tree.

But since all the old rules hold, it's fine to just start with where we ended last time.

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|----|---|----------------------------|
| 1. | $\mathbb{F} 1, \Diamond \Box A \rightarrow A$ ✓ | Assumption |
| 2. | $\mathbb{T} 1, \Diamond \Box A$ ✓ | $\rightarrow \mathbb{F} 1$ |
| 3. | $\mathbb{F} 1, A$ | $\rightarrow \mathbb{F} 1$ |
| 4. | $\mathbb{T} 1.1, \Box A$ ✓ | $\Diamond \mathbb{T} 2$ |
| 5. | $\mathbb{T} 1.1, A$ | $\mathbb{T} \Box 4$ |

Here's where we were.

1.	$\mathbb{F} 1, \Diamond \Box A \rightarrow A$ ✓	Assumption
2.	$\mathbb{T} 1, \Diamond \Box A$ ✓	$\rightarrow \mathbb{F} 1$
3.	$\mathbb{F} 1, A$	$\rightarrow \mathbb{F} 1$
4.	$\mathbb{T} 1.1, \Box A$ ✓	$\Diamond \mathbb{T} 2$
5.	$\mathbb{T} 1.1, A$	$\mathbb{T} \Box 4$
6.	$\mathbb{T} 1, A$	$\mathbb{B} \Box 4$
	⊗	

By applying the distinctive rule for **B** we get A being true at the one place we've said it can't be true. So the tree closes, and in **KTb** this is valid.